1 Model

We develop a static general equilibrium model of multinational production in which the decision maker is the business group, identified with its global ultimate parent. A group can operate plants in several countries and, when it sets quantities, takes into account that its own plants compete with one another: expanding one plant cannibalizes the sales of the others. Its markup therefore depends on the group's total share of the market rather than on the share of any single plant. Each group is owned by households in one or more countries, and its profits are paid to those owners wherever they are earned. Ownership shares are defined group by group, so two groups headquartered in the same country can have different owners. The baseline takes as given which plants each group operates and which markets they serve; Section 1.4 lets groups choose them. Formal statements and proofs are in the Appendix, and a table of notation closes the document.

1.1 Model Set-Up

Agents. There is a finite set of countries $\mathcal{N} = \{1, \dots, N\}$ and a finite set of sectors $\mathcal{K}$. Each country n is inhabited by a representative household that supplies L_n units of labor inelastically and owns shares in groups. Production takes place in plants, and every plant belongs to exactly one group $g \in \mathcal{G}$, which we identify with its global ultimate parent; a firm that is not part of a larger group is a group with a single plant. Each group has a headquarters country $h(g)$, operates in a single sector $k(g)$,[1] and owns a finite set of plants $\mathcal{P}_g$ (in Section 1.4, the plants it may operate). We index plants by i and write $g(i)$ and $\ell(i)$ for the group and the location of plant i. Each plant produces one differentiated variety, which it can sell in several countries.

Representative household. The household in country n has Cobb–Douglas preferences across sectors and CES preferences within them. Let $\mathcal{A}_{nk}$ be the set of plants that sell sector-k goods in country n, and $q_{i,n}$ the quantity of plant i's variety sold in n. Then

$$U_n = \prod_{k \in \mathcal{K}} C_{nk}^{\beta_k}, \qquad C_{nk} = \Big[\sum_{i \in \mathcal{A}_{nk}} q_{i,n}^{(\sigma_k - 1)/\sigma_k} \Big]^{\sigma_k/(\sigma_k - 1)}, \tag{1}$$

where $\beta_k > 0$, $\sum_k \beta_k = 1$, and $\sigma_k > 1$ is the elasticity of substitution between varieties of the same sector. The household spends the share β_k of its income X_n, defined in (5), on sector k. Writing E_{nk} for spending on sector k in n, demand is

$$q_{i,n} = p_{i,n}^{-\sigma_k} \, P_{nk}^{\sigma_k - 1} E_{nk}, \qquad P_{nk} = \Big[\sum_{i \in \mathcal{A}_{nk}} p_{i,n}^{1-\sigma_k} \Big]^{1/(1-\sigma_k)}, \tag{2}$$

where P_{nk} is the sector's price index. Welfare is real income X_n/P_n, where $P_n = \prod_k (P_{nk}/\beta_k)^{\beta_k}$ is the cost of one unit of utility.[2]

[1] This restriction is without loss. Marginal costs are constant, so a group's decisions separate across markets (Section 1.2), and a group active in several sectors behaves like several single-sector groups with the same owners.

[2] With Cobb–Douglas preferences across sectors, a sector's spending does not expand when its price index falls. A CES upper tier with elasticity $\eta \in (1, \sigma_k)$ adds that margin. It changes the markup in (7) to $1/\mu(S) = 1 - (1 - S)/\sigma_k - S/\eta$, and Proposition 2 still holds with this markup. Sector spending then depends on sector prices, however, and the general equilibrium results of Section 1.3 would have to be derived again.

Production technology. Labor is the only factor of production. A plant located in $\ell = \ell(i)$ produces with constant returns, and its unit cost is

$$a_i = \frac{w_\ell}{\varphi_i}, \tag{3}$$

where w_ℓ is the wage in ℓ and $\varphi_i > 0$ the plant's productivity.[3]

Head-office capability. Productivity can combine a component of the location, a component of the plant, and a component common to all plants of the group. The group component captures head-office services such as management, technology and brands (Antràs, 2003; Antràs and Helpman, 2004). These services are a non-rival capability of the group that all of its plants share, so they enter productivity rather than the cost of inputs.[4] A plant may also be less productive when it operates away from its group's headquarters (Ramondo and Rodríguez-Clare, 2013; Head and Mayer, 2019). The model uses only the resulting productivity $\varphi_i > 0$ of each plant: no result in this section depends on how φ_i splits into these components, and their parametric form is set in the quantification.

Trade. Shipping goods from ℓ to n entails an iceberg cost $d_{\ell n} \geq 1$, with $d_{nn} = 1$. Trade costs run from the country where the good is produced, not from headquarters, so a single plant can serve several destinations, including third markets, as an export platform (Tintelnot, 2017). Country n levies a tariff $t_{\ell nk} \geq 0$ on sector-k goods produced in ℓ, with $t_{nnk} = 0$. Tariffs depend on where a good is produced, not on who owns the plant that produces it, so a country's tariffs also fall on the foreign affiliates of the groups its residents own. We model the tariff as a proportional wedge on the cost of delivering the good. The delivered marginal cost of plant i in market (n, k) is

$$c_{i,n} = a_i \, d_{\ell n} \, (1 + t_{\ell nk}), \qquad \ell = \ell(i), \tag{4}$$

and country n collects $t_{\ell nk} \, a_i d_{\ell n} q_{i,n}$ on the sales of plant i.[5]

Multinational corporations. A *market* is a destination–sector pair (n, k). In the baseline, each plant serves a given set of destinations, so the set $\mathcal{A}_{nk}$ of plants that serve market (n, k) is given. Plant i is a foreign affiliate if it is located outside its group's headquarters country, $\ell(i) \neq h(g(i))$. A group with at least one foreign affiliate is a multinational; any other group is domestic, and a domestic group with a single plant is a stand-alone firm. A multinational can serve a market from a plant at home, from a plant in the destination, from a plant in a third country, or from several at once. We assume that every market is served by at least two groups, that is, $\mathcal{P}_g \cap \mathcal{A}_{nk} \neq \emptyset$ for at least two groups g: under Cournot competition with Cobb–Douglas spending across sectors, a group that served a market alone would face no limit on its markup. Section 1.4 lets groups choose which plants serve each market, so that multinational status becomes an equilibrium outcome.

[3] We do not include intermediate inputs. With an input–output structure, a tariff also raises the cost of the input bundle, and the existence and uniqueness results below require additional conditions.

[4] The common alternative treats head-office services as labor bought at the parent's wage, so that a plant abroad pays part of its costs at a foreign wage. This links factor markets across countries and can break the gross-substitutes property (Section 1.3) on which uniqueness of wages rests.

[5] A duty on the exporter's price yields the same prices when all the plants a group uses to serve n face the same tariff, but it also taxes the markup, so it moves that part of the duty from the group to the treasury. When the plants face different tariffs, it makes the group's markup differ across its plants, and the property that a group's markup depends only on its total share (Section 1.2) is lost.

Profit repatriation. Country n owns a share $\theta_{gn} \geq 0$ of group g, with $\sum_n \theta_{gn} = 1$. Groups maximize their own profits whoever owns them: ownership determines where profits are spent, not how groups price. The profit of a group, $\pi_g = \sum_n \pi_{g,n}$, where $\pi_{g,n}$ is its profit in market $(n, k(g))$ (Section 1.2), is paid to its owners in proportion to θ_{gn}, wherever it is earned. Tariff revenue is rebated lump sum to the household of the country that collects it. Income in country n is

$$X_n = w_n L_n + \sum_{g \in \mathcal{G}} \theta_{gn}\, \pi_g + TR_n, \qquad TR_n = \sum_k \sum_{i \in \mathcal{A}_{nk}} t_{\ell(i)nk}\, a_i\, d_{\ell(i)n}\, q_{i,n}. \tag{5}$$

A natural benchmark assigns each group to the country of its ultimate parent, $\theta_{gn} = \mathbf{1}\{n = h(g)\}$; nothing below requires this form. Because income includes the profits of the groups a country owns, wherever they produce and sell, trade need not balance country by country.

Timing. The model is static. Plants, productivities, trade costs, tariffs and ownership shares are given. Groups simultaneously choose the quantities their plants sell in each market, taking wages and market sizes as given and taking into account their effect on the market's price index, and households choose their consumption.

1.2 Market Structure

Fix a market (n, k) and suppress its indices: we write σ, E and P for σ_k, E_{nk} and P_{nk}, c_i for the delivered cost $c_{i,n}$, $T_g = \mathcal{P}_g \cap \mathcal{A}_{nk}$ for the plants of group g that serve the market, and π_g for the group's profit $\pi_{g,n}$ in the market.

Demand. Let $A \equiv \sum_{i \in \mathcal{A}} p_i^{1-\sigma} = P^{1-\sigma}$. By (2), the revenue share of plant i and the share of group g are

$$s_i = \frac{p_i q_i}{E} = \frac{p_i^{1-\sigma}}{A}, \qquad S_g = \sum_{i \in T_g} s_i, \qquad \sum_g S_g = 1. \tag{6}$$

The aggregate A summarizes how competitive the market is: it is large when rivals charge low prices, and then any given price wins a small share.

Competition. Groups are the strategic agents. Each group chooses the quantities of all of its plants that serve the market, taking as given the quantities of other groups and the level of spending E. It maximizes its profit $\pi_g = \sum_{i \in T_g} (p_i - c_i) q_i$.

The first-order conditions of group g (Proposition 1 in Appendix A.1) give $p_i = \mu(S_g)\, c_i$ for every plant $i \in T_g$, with

$$\mu(S) = \frac{\sigma}{(\sigma - 1)(1 - S)}. \tag{7}$$

This is the markup of Atkeson and Burstein (2008), evaluated at the share of the group rather than of the plant.[6] It rises from $\sigma/(\sigma - 1)$ for a negligible group and is unbounded as the share tends to one.

[6] Under price competition the markup is $\varepsilon(S)/[\varepsilon(S) - 1]$ with $\varepsilon(S) = \sigma - (\sigma - 1)S$. It is also common to the group's plants and increasing in its share, so the structure below carries over, but profits are no longer quadratic in the share.

Cannibalization across a group's plants. When a firm expands output, the sector price index falls and every variety in the market loses sales. A single-plant firm takes into account the sales it loses itself, so its marginal revenue is $\frac{\sigma-1}{\sigma}p_i(1-s_i)$, which gives the markup of Atkeson and Burstein (2008) at its own share. A group also counts the sales its other plants lose, so the marginal revenue of plant i is $\frac{\sigma-1}{\sigma}p_i(1-S_g)$ (Appendix A.1), which gives (7): its plants cannibalize one another, and the loss grows with the group's share. Two implications follow. First, all plants of a group charge the same markup in a given market, and differences in their costs show up in quantities rather than in markups. Second, market power is a property of the group: concentration should be measured at the level of groups, and each plant's markup depends on its group's share, which exceeds its own. Within a group, (6) and the common markup give

$$\frac{s_i}{S_g} = \frac{c_i^{1-\sigma}}{K_g}, \qquad K_g \equiv \sum_{i \in T_g} c_i^{1-\sigma}. \tag{8}$$

We call K_g the *strength* of the group in the market: an index of its cost efficiency that rises with the number of plants serving the market and with how cheaply each delivers. Substituting $p_i = \mu(S_g)c_i$ into (6),

$$\psi(S_g) = \frac{K_g}{A}, \qquad \psi(S) \equiv S\,\mu(S)^{\sigma-1} = \Big(\frac{\sigma}{\sigma-1}\Big)^{\sigma-1}\frac{S}{(1-S)^{\sigma-1}}. \tag{9}$$

The map ψ is strictly increasing, so $S_g = \psi^{-1}(K_g/A)$: the share of a group depends only on its strength relative to the competitiveness of the market. However many plants a group runs, and in however many countries, its position in the market is summarized by one number, K_g, and all it needs to know about its rivals is A. The equilibrium of the market is the value of A at which shares add up to one,

$$\sum_g \psi^{-1}\Big(\frac{K_g}{A}\Big) = 1. \tag{10}$$

Profits. Since $c_i q_i = p_i q_i/\mu(S_g)$, the profit of group g is $\pi_g = E\,S_g\,m(S_g)$, where $m(S) \equiv 1 - 1/\mu(S)$ is the Lerner index. By (7) the Lerner index is linear in the share, $m(S) = (1+\kappa S)/\sigma$ with $\kappa \equiv \sigma - 1$, so that

$$\pi_g = \frac{E}{\sigma}(S_g + \kappa S_g^2). \tag{11}$$

Profits rise more than proportionally with the share, because a larger group both sells more and earns a higher margin on each dollar of sales. The curvature κ would be zero with constant markups. Because $\kappa > 0$, the profit income a country receives through (5) depends on which groups it owns, not only on its average ownership share: owning the large groups is worth more than owning the same share of sales spread over small ones.

1.3 General Equilibrium

Market sizes. Without intermediate inputs, all spending is final spending by households, so spending on market (n,k) is

$$E_{nk} = \beta_k X_n. \tag{12}$$

Labor market. Let $V_{i,n'} \equiv a_i d_{\ell(i)n'} q_{i,n'}$ be the cost of plant i in serving country n', exclusive of tariffs. Since each price is the group's markup times delivered cost, $V_{i,n'} = r_{i,n'}/[\mu_{g(i)}(1+t_{\ell(i)n'k})]$,

where $r_{i,n'}$ is the plant's revenue in that market, $\mu_{g(i)}$ its group's markup there, and k the plant's sector. All of this cost is paid to labor where the plant is located, so labor markets clear when

$$w_n L_n = \sum_{k \in \mathcal{K}} \sum_{n' \in \mathcal{N}} \sum_{i \in \mathcal{A}_{n'k}:\, \ell(i)=n} V_{i,n'}. \tag{13}$$

Equilibrium. An equilibrium consists of wages $\{w_n\}$, incomes $\{X_n\}$, and prices and quantities $\{p_{i,n}, q_{i,n}\}$ such that:

1. households maximize (1), so quantities satisfy (2);
2. in every market, each group's prices satisfy its first-order conditions, (7);
3. incomes satisfy (5), with market sizes (12);
4. labor markets clear, (13).

Prices, costs and incomes are homogeneous of degree one in wages, and shares are homogeneous of degree zero, so wages are determined up to scale. At any wages, once incomes satisfy (5), the value of labor demanded across all countries equals the value of labor supplied (Walras' law, Appendix B.2), so one labor-market condition is implied by the others. We set $w_1 = 1$.

Existence and uniqueness. We establish the results layer by layer: the equilibrium of a single market, incomes given wages, and wages. The formal statements and their proofs are in Appendices A and B.

For any positive delivered costs, a market has exactly one equilibrium (Proposition 2). The proof uses (10). Each $\psi^{-1}(K_g/A)$ falls as A rises, from one when A is near zero to zero when A is large. The sum of shares therefore falls strictly from the number of groups, at least two, to zero, and equals one exactly once. Each group's profit is strictly concave in its own quantities, so the first-order conditions identify the unique best response.

Given wages, incomes are the unique solution of the linear system (5), (12), and $X_n \geq w_n L_n$ for every n (Proposition 3). Every share and markup is then known, profits and tariff revenue are linear in market sizes, and market sizes are linear in incomes. Part of every dollar spent in a market is paid to labor, so less than a dollar returns to households as profits and tariff revenue; this makes the system's solution unique. Since profits and tariff revenue are non-negative, income is at least labor income.

Wages that clear every labor market exist provided no set of countries is cut off from the rest: for every proper subset Z of countries, some plant located in Z serves a market of a country outside Z (Proposition 4). The condition rules out a set of countries that sells nothing to the rest of the world, whose wages relative to the others would then be undetermined.

Uniqueness of wages is the most delicate layer. Let $W_n(w)$ be the right side of (13), the wage bill of country n at wages w, and $M_{nj} \equiv \partial \ln W_n / \partial \ln w_j$ its elasticity with respect to country j's wage. Homogeneity makes every row of M sum to one. Wages are gross substitutes when $M_{nj} > 0$ for $n \neq j$: a rise in any other country's wage raises country n's wage bill. Let $b^n_{i,n'}$ be the share of country n's wage bill that plant i pays out of its sales in country n', and

$$\bar{E}^{(j)}_n \equiv \sum_{i,n'} b^n_{i,n'} \frac{\partial \ln E_{n'k(i)}}{\partial \ln w_j}, \tag{14}$$

the average response of the markets in which country n's plants sell. Theorem 1 gives sufficient conditions for uniqueness. Let D be a set of wage vectors that is convex in logarithms: the straight line in logarithms between any two of its points stays in D. If at every $w \in D$ (i) no group holds

more than half of any market and (ii) $\bar{E}_n^{(j)} > 0$ for all $n \neq j$, then $M_{nj} \geq \bar{E}_n^{(j)} > 0$ for $n \neq j$ throughout D, and D contains at most one equilibrium wage vector up to scale.

The bound $M_{nj} \geq \bar{E}_n^{(j)}$ says that a rise in a foreign wage can lower country n's wage bill only by shrinking the markets in which its plants sell. Plants located in n do not become more expensive when w_j rises, so every reallocation of sales between plants and between groups favors country n. Condition (i) is what signs the effect of variable markups on labor demand. With gross substitutes, a second equilibrium is ruled out by the classical argument of Arrow, Block, and Hurwicz (1959). Condition (ii) can fail. A country whose groups lose profits when a foreign wage rises spends less, and if the markets where country n sells shrink enough, its wage bill can fall. That income effect is the ownership channel the model is built to study. Both conditions are inequalities in objects the model computes, and we report them with the quantification.

Solution algorithm. The layers above map into nested loops. Given wages, each market is solved for A from (10), and incomes come from one linear solve. Wages are updated by the damped tâtonnement $w_n \leftarrow w_n^{1-\lambda}(W_n(w)/L_n)^{\lambda}$, with $\lambda \in (0,1)$, which raises a country's wage when its labor is in excess demand and whose fixed points are the equilibrium wages. When labor demand is elastic, $M_{nn} < 0$, an undamped update can overshoot; a λ with $\lambda(1 - M_{nn}) < 1$ avoids it.

1.4 Extension: Endogenous Entry

The baseline takes as given which plants each group operates and which markets they serve. This subsection lets groups choose them, so that which affiliates exist, and where each of them sells, is an equilibrium outcome.

Fixed costs and plant sets. Each group now has a set of potential plants $\mathcal{P}_g$. A plant that serves market (n,k) pays a fixed cost

$$F_{i,n} = f_k \, w_\ell \, f_{\ell n}^d, \tag{15}$$

in units of labor of the country ℓ where it is located, with $f_k > 0$ a sector component and $f_{\ell n}^d \geq 1$ a bilateral component that is larger when serving n from ℓ is more costly. The cost is incurred once per market served, whatever the quantity sold, and does not depend on who owns the plant. Because it is paid in labor, it appears as wage income and leaves the model homogeneous of degree one in wages. Every fixed cost is specific to a plant–market pair, so a group's problem separates across destinations: in each market $(n, k(g))$ it chooses the subset $T_{g,n} \subseteq \mathcal{P}_g$ of its plants that serve n.[7] This is a simplified version of the multi-plant location problem of Yang (2023).

The entry problem. A group's plant set matters for its profit only through its strength. For a set of plants $T \subseteq \mathcal{P}_g$, let $K(T) = \sum_{i \in T} c_i^{1-\sigma}$ be its strength and $\Phi(T) = \sum_{i \in T} F_{i,n}$ its fixed cost. Given A, serving the market with T yields

$$U_g(T;A) = \Omega\big(K(T);A\big) - \Phi(T), \qquad \Omega(K;A) \equiv \frac{E}{\sigma}(S + \kappa S^2)\Big|_{S=\psi^{-1}(K/A)}. \tag{16}$$

Because Ω is strictly increasing in K, a set can be optimal only if no other set delivers at least as much strength at no greater fixed cost. These efficient sets form a chain, ordered by strength and by cost together, so the group's choice amounts to how far to climb a ladder that runs from

[7] A plant-level setup cost would link a group's choices across markets and turn its problem into a combinatorial one over locations and markets jointly (Tintelnot, 2017).

small and cheap to large and expensive. When a group's plants pay the same fixed cost to serve the market, each rung adds the plant with the next-lowest delivered cost, and the group's choice is a cutoff on its own list of plants, the within-group analogue of the cutoff in Gaubert and Itskhoki (2021).

Entry equilibrium. Given delivered costs, fixed costs and spending, an entry equilibrium in a market is a collection of plant sets $\{T_g\}$ and an aggregate A such that each T_g maximizes $U_g(\cdot\,; A)$ over subsets of $\mathcal{P}_g$, and A solves (10) with $K_g = K(T_g)$. Since (10) has no solution when fewer than two groups serve the market, every entry equilibrium has at least two. Groups evaluate entry at the prevailing degree of competition, as in a free-entry condition, while taking into account their effect on A when they set quantities. This notion turns each group's entry problem into the choice of its strength K given the aggregate A; Appendix C.4 discusses the alternative in which groups also anticipate their effect on A when they enter.

Theorem 2 in Appendix C establishes three properties when $\sigma \geq 2$, a bound the proof uses to sign one term (Appendix C.2). First, $\Omega(K; A)$ is strictly concave in K at every share: strength has diminishing returns for the group. This is cannibalization at the entry margin: at a given A, a new plant takes part of its sales from the group's own plants, so the group's share rises less than proportionally with its strength. The higher markup the group then earns on all of its sales works the other way, and part (a) shows that cannibalization dominates. Second, the cross partial derivative Ω_{KA} is negative whenever the group's share is at most one half: strength is worth less in a more competitive market. Third, a market has at most one entry equilibrium, provided that, for any two candidate equilibria, no group would hold more than half of the market if it operated the plants it chooses in one of them while facing the aggregate of the other, and that no group is indifferent between two rungs of its ladder. The first two properties imply that each group's optimal strength does not rise when A rises, while A rises with every group's strength, so two equilibria with different aggregates cannot both exist. No order of entry is imposed, and groups still internalize competition among their own plants. The hypothesis is the same one-half condition as in Theorem 1.

General equilibrium with entry. Fixed costs are paid to labor where the plant is located, so they are added to the right side of (13), and profits distributed through θ are net of them. For given plant sets, the results of Section 1.3 carry over with two changes. Profits net of fixed costs can be negative, so income can fall below labor income, and the existence proof (Appendix B.2) uses a lower bound on income: existence of wages requires fixed costs small enough that every country's income is at least a fixed fraction of its labor income. In the demand term (14), payments for fixed costs enter with a zero response, because a fixed cost paid in country n does not change when another country's wage rises. Two limits are new. First, plant sets change discretely with wages, so uniqueness at each layer does not make the joint equilibrium of wages and plant sets unique. Second, entry is chosen in whole plants, so an equilibrium in which no group wishes to change its plant set need not exist. The solution algorithm adds an outer loop over plant sets: at the equilibrium wages for the current sets, the groups with the largest gains from changing their sets do so. This changes the path of the algorithm, not its fixed points, and imposes no order of entry. When the loop does not reach a configuration in which no group wishes to move, it returns the one with the fewest such groups and reports how many there are.

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Appendix

A The Market Game

A.1 Markups.

Proposition 1 (Markups)**.** *The first-order conditions of group g are $p_i = \mu(S_g)\, c_i$ for every $i \in T_g$, with μ given by* (7)*.*

Proof. Fix a market. Let $\gamma \equiv (\sigma-1)/\sigma$ and $Q \equiv (\sum_i q_i^\gamma)^{1/\gamma}$, the quantity of the sectoral composite. Since $PQ = E$, inverse demand is $p_i = E\, Q^{-\gamma} q_i^{\gamma-1}$, so $s_i = q_i^\gamma/Q^\gamma$ and group g's revenue is $E\, Q^{-\gamma} \sum_{j\in T_g} q_j^\gamma = E\, S_g$. Using $\partial \ln Q/\partial q_i = q_i^{\gamma-1}/Q^\gamma = s_i/q_i$, the derivative of the group's revenue with respect to q_i is

$$E\, Q^{-\gamma}\Big[\gamma\, q_i^{\gamma-1} - \gamma\, \frac{q_i^{\gamma-1}}{Q^\gamma} \sum_{j\in T_g} q_j^\gamma\Big] = \gamma\, p_i\, (1 - S_g).$$

The second term in the bracket is the sales that expanding plant i takes from all of the group's plants, plant i included; for a single-plant firm only the term $j = i$ remains, which gives the factor $1 - s_i$. Setting marginal revenue equal to c_i gives (7). □

A.2 One market.

Proposition 2 (One market)**.** *For any positive delivered costs, a market has exactly one equilibrium.*

Proof. Best responses. Given rivals' quantities, group g's profit is $E\, X_g/(X_g + B_g) - \sum_{i\in T_g} c_i q_i$, where $X_g = \sum_{i\in T_g} q_i^\gamma$ and $B_g > 0$ is the same index for rivals. The function $X \mapsto X/(X+B)$ is strictly increasing and strictly concave, with derivatives $B/(X+B)^2 > 0$ and $-2B/(X+B)^3 < 0$, and X_g is strictly concave in $(q_i)_{i\in T_g}$. A strictly increasing, strictly concave function of a strictly concave function is strictly concave, and cost is linear, so profit is strictly concave in the group's own quantities. Revenue is at most E, so profit falls without bound as any quantity grows, and a maximum exists. Since $\partial X_g/\partial q_i \to \infty$ as $q_i \to 0$, the optimum is interior and is the unique solution of the first-order conditions. *The share map.* By (9), $\psi(0) = 0$, ψ is continuous and strictly increasing on $[0, 1)$, and $\psi(S) \to \infty$ as $S \to 1$, so it has a continuous, strictly increasing inverse defined on $[0, \infty)$. *The aggregate.* By Proposition 1 and (9), an equilibrium is an A solving (10), and conversely any such A generates prices $p_i = \mu(S_g)c_i$ that satisfy every group's first-order conditions, and hence are best responses by the concavity above, and that are consistent with A, since $\sum_i p_i^{1-\sigma} = \sum_g \mu(S_g)^{1-\sigma} K_g = A \sum_g S_g = A$. The left side of (10), $\mathcal{S}(A)$, is continuous and strictly decreasing. As $A \to \infty$ every share tends to zero, and as $A \to 0$ every share tends to one, so $\mathcal{S}$ tends to the number of groups, which is at least two. Hence $\mathcal{S}(A) = 1$ has exactly one solution. □

B General Equilibrium

B.1 Incomes.

Proposition 3 (Incomes)**.** *Given wages, incomes are the unique solution of the linear system* (5)*,* (12)*, and $X_n \geq w_n L_n$ for every n.*

Proof. Given wages, delivered costs are known, and by Proposition 2 so are all shares and markups. In market (n', k), group g's profit is $E_{n'k} S_{g,n'} m(S_{g,n'})$ and tariff revenue is $\tau_{n'k} E_{n'k}$, with $\tau_{n'k} = \sum_{i \in \mathcal{A}_{n'k}} t_i s_i / [\mu_{g(i)}(1+t_i)]$ and t_i the tariff on plant i. Substituting (12) into (5),

$$X_n = w_n L_n + \sum_{n'} \Gamma_{nn'} X_{n'}, \qquad \Gamma_{nn'} = \sum_k \beta_k \Big[\sum_{g:\, k(g)=k} \theta_{gn} S_{g,n'} m(S_{g,n'}) + \mathbf{1}\{n' = n\}\, \tau_{nk} \Big] \geq 0,$$

where the inequality uses non-negative tariffs, so that $\tau_{nk} \geq 0$. Using $\sum_n \theta_{gn} = 1$, the column of Γ for n' sums to $\sum_k \beta_k [\sum_g S_{g,n'} m(S_{g,n'}) + \tau_{n'k}]$. Per dollar of spending in a market, revenue splits into profit, tariff revenue and labor cost,

$$\sum_g S_g m(S_g) + \tau + \sum_i \frac{s_i}{\mu_{g(i)}(1+t_i)} = 1, \tag{B.1}$$

and the last term is positive. So every column of Γ sums to less than one, the largest eigenvalue of Γ in absolute value is below one, and $X = (I - \Gamma)^{-1} wL = \sum_{s \geq 0} \Gamma^s wL \geq wL$ is the unique solution. □

B.2 Existence of wages.

Proposition 4 (Existence of wages)**.** *Suppose that for every proper subset Z of countries, some plant located in Z serves a market of a country outside Z. Then there exist wages that clear every labor market.*

Proof. Normalize wages to the unit simplex and let $z_n(w) = W_n(w)/w_n - L_n$ be excess demand for labor. By Propositions 2–3, z is continuous on the interior of the simplex and homogeneous of degree zero. It satisfies Walras' law $\sum_n w_n z_n(w) = 0$. Adding (5) across countries, world income is $\sum_n w_n L_n$ plus world profits plus world tariff revenue. By (12), world spending equals world income, and by (B.1) it also equals $\sum_n W_n$ plus world profits plus world tariff revenue. Hence $\sum_n W_n = \sum_n w_n L_n$. Excess demand is bounded below by $-L$. It remains to check the boundary condition of Mas-Colell, Whinston, and Green (1995, Proposition 17.C.1): if $w^m \to \bar{w}$, where $Z = \{n : \bar{w}_n = 0\}$ is non-empty, then $\max_n z_n(w^m) \to \infty$.

A plant's delivered cost is proportional to the wage where it is located. So the costs of plants located in Z go to zero, while all other costs stay in a compact subset of $(0, \infty)$. By hypothesis, some plant located in Z serves a market (n', k) with $n' \notin Z$. By Proposition 3, $X_{n'} \geq w_{n'} L_{n'}$, so $E_{n'k} = \beta_k X_{n'}$ stays bounded away from zero. For m large, the lowest-cost plant i in that market is located in some $\ell \in Z$; since there are finitely many plants, we can pass to a subsequence along which ℓ, i and its group g do not change. Let J be the number of plants in the market. Its cost is $V_i = E_{n'k} s_i / [\mu_g(1+t_i)]$, all of it paid to labor in ℓ, so it suffices to show that $V_i / w_\ell \to \infty$.

If $S_g \leq \frac{1}{2}$, then $\mu_g \leq 2\sigma/(\sigma - 1)$. Every price is at least c_i, since markups exceed one, so $P^{1-\sigma} \leq J c_i^{1-\sigma}$ and $s_i = (\mu_g c_i / P)^{1-\sigma} \geq \mu_g^{1-\sigma} / J$. Then V_i is bounded away from zero while $w_\ell \to 0$.

If $S_g > \frac{1}{2}$, every rival h has $S_h \leq 1 - S_g < \frac{1}{2}$, so $(1 - S_h)^{1-\sigma} \leq 2^{\sigma - 1}$ and $1/S_g < 2$. By (9),

$$\frac{K_h}{K_g} = \frac{\psi(S_h)}{\psi(S_g)} = \frac{S_h (1 - S_h)^{1-\sigma}}{S_g (1 - S_g)^{1-\sigma}} \leq 2^\sigma (1 - S_g)^\sigma,$$

so $1 - S_g \geq \frac{1}{2} (K_h / K_g)^{1/\sigma}$. Rival strength K_h is bounded below, because all costs are bounded above. Plant i is the cheapest in its group, so $K_g \leq J c_i^{1-\sigma}$ and $s_i \geq S_g / J > 1/(2J)$. With $1/\mu_g = (\sigma - 1)(1 - S_g)/\sigma$, this gives $V_i \geq c\, c_i^{(\sigma-1)/\sigma}$ for some $c > 0$, and since c_i is proportional to w_ℓ, V_i / w_ℓ grows like $w_\ell^{-1/\sigma}$.

In either case $z_\ell \to \infty$, and the result follows from Mas-Colell, Whinston, and Green (1995, Proposition 17.C.1). □

B.3 A decomposition of labor payments. Raise wages, holding tariffs and trade costs fixed, and let hats denote log changes. A plant's delivered cost changes by $\hat{c}_i = \hat{w}_{\ell(i)}$. Its cost in a market is $V_i = r_i/[\mu_g(1+t_i)]$, so $\hat{V}_i = \hat{E} + \hat{s}_i - \hat{\mu}_g$ with $\hat{s}_i = (1-\sigma)(\hat{c}_i + \hat{\mu}_g - \hat{P})$ and $\hat{\mu}_g = \varepsilon_\mu \hat{S}_g$, where $\varepsilon_\mu(S) \equiv S\mu'(S)/\mu(S) = S/(1-S)$ is the elasticity of the markup. Let $\hat{c}_g = \sum_{i \in T_g}(s_i/S_g)\hat{c}_i$ be the group's average cost change and

$$\Lambda_g \equiv \frac{1}{1+(\sigma-1)\varepsilon_\mu(S_g)} = \frac{1-S_g}{1+(\sigma-2)S_g}, \qquad \varpi_g \equiv \frac{S_g\Lambda_g}{\sum_h S_h\Lambda_h}.$$

By (8), $\hat{K}_g = (1-\sigma)\hat{c}_g$, and $\hat{A} = (1-\sigma)\hat{P}$. Differentiating $\psi(S_g) = K_g/A$, with $d\ln\psi/d\ln S = 1/\Lambda_g$, gives $\hat{S}_g = -(\sigma-1)\Lambda_g(\hat{c}_g - \hat{P})$. Shares sum to one, so $\sum_g S_g\hat{S}_g = 0$, which gives $\hat{P} = \sum_g \varpi_g \hat{c}_g$. Using $1 = \Lambda_g[1+(\sigma-1)\varepsilon_\mu]$,

$$\hat{V}_i = \hat{E} - (\sigma-1)(\hat{c}_i - \hat{c}_g) - \chi_g(\hat{c}_g - \hat{P}), \qquad \chi_g \equiv (1-\varepsilon_\mu)(\sigma-1)\Lambda_g = \frac{(\sigma-1)(1-2S_g)}{1+(\sigma-2)S_g}. \quad \text{(B.3)}$$

The first term is the growth of the market, the second the reallocation of sales within the group, the third the reallocation between the group and its rivals. The coefficient χ_g lies in $[0, \sigma-1]$ exactly when $S_g \le \frac{1}{2}$.

B.4 Uniqueness of wages.

Theorem 1 (Wages)**.** *Let D be a set of wage vectors that is convex in logarithms. Suppose that at every $w \in D$ (i) no group holds more than half of any market and (ii) $\bar{E}_n^{(j)} > 0$ for all $n \neq j$. Then $M_{nj} \ge \bar{E}_n^{(j)} > 0$ for $n \neq j$ throughout D, and D contains at most one equilibrium wage vector up to scale.*

Proof. The bound. Raise w_j alone, $j \neq n$. Every plant paying country n is located in n, so $\hat{c}_i = 0$ for those plants. Every other plant has $\hat{c}_i \in \{0,1\}$, so $\hat{c}_g \in [0,1]$ and, by B.3, $\hat{P} = \sum_g \varpi_g \hat{c}_g \in [0,1]$. With $\hat{c}_i = 0$ the last two terms of (B.3) equal $(\sigma-1)\hat{c}_g - \chi_g(\hat{c}_g - \hat{P})$. If $\hat{c}_g \ge \hat{P}$, condition (i) gives $\chi_g \le \sigma - 1$ and the expression is at least $(\sigma-1)\hat{P} \ge 0$; otherwise both parts are non-negative, since $\chi_g \ge 0$. So $\hat{V}_i \ge \hat{E}$ for every plant paying country n, and weighting by $b^n_{i,n'}$, which sum to one, gives $M_{nj} \ge \bar{E}_n^{(j)}$. *Uniqueness.* The argument is classical (Arrow, Block, and Hurwicz, 1959); see also Mas-Colell, Whinston, and Green (1995, Section 17.F). Since M is homogeneous of degree zero in wages, conditions (i) and (ii) also hold at every rescaling of a point of D, and the set of these rescalings is still convex in logarithms. Let $w, w^* \in D$ be non-proportional equilibria. Scale w^* so that $w^* \le w$ with equality at some n_0, and move from w^* to w along the segment in logarithms. Every wage other than w_{n_0} weakly rises and at least one strictly rises, so W_{n_0} strictly rises while w_{n_0} does not. Then $z_{n_0}(w) > z_{n_0}(w^*) = 0$, a contradiction. □

C Entry

Theorem 2 (Entry)**.** *Let $\sigma \ge 2$.*

(a) $\Omega(K; A)$ is strictly concave in K at every share.

(b) $\Omega_{KA} < 0$ whenever the group's share is at most one half.

(c) Suppose that, for any two candidate equilibria, no group would hold more than half of the market if it operated the plants it chooses in one of them while facing the aggregate of the other, and that no group is indifferent between two rungs of its ladder. Then a market has at most one entry equilibrium.

Paragraphs C.1–C.3 prove the theorem; C.4 discusses the notion of equilibrium it uses.

C.1 Two elasticities. Let $G(S) \equiv \pi'(S)/\psi'(S)$, where $\pi(S) = E(S + \kappa S^2)/\sigma$, and define $\varepsilon_G = SG'/G$ and $\varepsilon_\psi = S\psi'/\psi$. Differentiating $\psi(S) = K/A$ gives $\partial S/\partial K = 1/(A\psi')$ and $\partial S/\partial A = -\psi/(A\psi')$. Hence $\Omega_K = G(S)/A > 0$, $\Omega_{KK} = G'(S)/(A^2\psi'(S))$, and

$$\Omega_{KA} = -\frac{1}{A^2}\Big[G'(S)\frac{\psi(S)}{\psi'(S)} + G(S)\Big] = -\frac{G(S)}{A^2\varepsilon_\psi(S)}\big[\varepsilon_G(S) + \varepsilon_\psi(S)\big].$$

So Ω is strictly concave in K if and only if $\varepsilon_G < 0$, and $\Omega_{KA} < 0$ if and only if $\varepsilon_G + \varepsilon_\psi > 0$.

C.2 Proof of Theorem 2(a)–(b). Set $E = 1$, $a \equiv 2(\sigma - 1)$ and $b \equiv \sigma - 2 \geq 0$. Then $\pi'(S) = (1 + aS)/\sigma$ and $\psi'(S) = C(1 - S)^{-\sigma}(1 + bS)$ with $C = [\sigma/(\sigma - 1)]^{\sigma-1}$, so

$$\varepsilon_G(S) = S\, f(S), \qquad f(S) = \frac{a}{1 + aS} - \frac{b}{1 + bS} - \frac{\sigma}{1 - S}.$$

At $S = 0$, $f(0) = a - b - \sigma = 0$. Since $a > b \geq 0$, $a/(1 + aS) > b/(1 + bS) \geq 0$, so $f'(S) = -a^2/(1 + aS)^2 + b^2/(1 + bS)^2 - \sigma/(1 - S)^2 < 0$. Hence $f < 0$ and $\varepsilon_G < 0$ on $(0, 1)$, which is (a). Next, $\varepsilon_\psi = (1 + bS)/(1 - S)$, and using $1 + bS - \sigma S = 1 - 2S$,

$$\varepsilon_G + \varepsilon_\psi = \frac{aS}{1 + aS} - \frac{bS}{1 + bS} + \frac{1 - 2S}{1 - S}.$$

The map $x \mapsto x/(1+x)$ is increasing and $a > b$, so the first two terms sum to a positive number; the third is non-negative when $S \leq \frac{1}{2}$. This is (b). At $S = \frac{1}{2}$ the expression equals $(\sigma-1)/\sigma-(\sigma-2)/\sigma = 1/\sigma$. □

C.3 Proof of Theorem 2(c). *Step 1: the ladder.* Since $\Omega_K > 0$, a set T is optimal only if no T' has $K(T') \geq K(T)$ and $\Phi(T') \leq \Phi(T)$ with one inequality strict. The remaining efficient sets can be ordered so that K and Φ rise together. *Step 2: single crossing.* Take $A < A'$ and two rungs with $K_1 < K_2$, and let $\Delta(x) = \int_{K_1}^{K_2} \Omega_K(K; x)\, dK - [\Phi_2 - \Phi_1]$ be the net gain from the larger rung at aggregate x. The share $\psi^{-1}(K/x)$ rises with K and falls with x, so on $[K_1, K_2] \times [A, A']$ it is largest at (K_2, A). If $\psi^{-1}(K_2/A) \leq \frac{1}{2}$, part (b) gives $\Omega_{KA} < 0$ throughout, and Δ is strictly decreasing on $[A, A']$. *Step 3: the aggregate.* Given strengths, the aggregate $A(\{K_g\})$ solving (10) is strictly increasing in each K_g: raising K_j at a given A raises the left side, which is strictly decreasing in A. *Step 4.* Suppose $(A, \{K_g\})$ and $(A', \{K'_g\})$ are equilibria with $A < A'$, and suppose $K'_g > K_g$ for some g. Apply Step 2 with $K_1 = K_g$ and $K_2 = K'_g$: by the hypothesis, group g would hold at most half of the market with the plants it chooses at A' while facing A, so $\psi^{-1}(K'_g/A) \leq \frac{1}{2}$ and Δ is strictly decreasing on $[A, A']$. Optimality at A' gives $\Delta(A') \geq 0$, hence $\Delta(A) > 0$, so at A the group strictly prefers K'_g to K_g, contradicting optimality at A. Hence $K'_g \leq K_g$ for every g (the single-crossing argument of Milgrom and Shannon, 1994), and by Step 3, $A' = A(\{K'_g\}) \leq A(\{K_g\}) = A$, a contradiction. Equilibria with the same aggregate have the same best responses, and with no ties the same plant sets. □

C.4 The notion of entry equilibrium. The entry equilibrium of Section 1.4 evaluates every alternative set at the prevailing aggregate A. In a two-stage game in which the market is re-solved after a deviation, a group that adds a plant anticipates that A rises, which lowers the value of the deviation; the two notions coincide for groups with negligible shares and can differ for large ones. Theorem 2 applies to the first notion. Existence in pure strategies is not guaranteed under either notion: a group's choice is discrete, so its optimal strength jumps with A, and (10) may have no solution with each group on a rung.

Notation

Symbols are grouped by topic, and the last column gives the equation or section where each is defined. A subscript n or n' denotes a destination country, ℓ the country where a plant is located, k a sector, g, h or j a group, and i a plant. Within a single market the market's indices are dropped. Symbols used only inside one proof, such as γ, Q and G, are defined there.

Symbol	Meaning	Where
Agents and sets		
$\mathcal{N}$, N	set of countries; number of countries	Agents
$\mathcal{K}$	set of sectors	Agents
$\mathcal{G}$	set of groups (a group is a global ultimate parent and all its plants)	Agents
$h(g)$, $k(g)$	headquarters country and sector of group g	Agents
$\mathcal{P}_g$	plants owned by group g (potential plants in the extension)	Agents
$g(i)$, $\ell(i)$	group and location of plant i	Agents
$\mathcal{A}_{nk}$	plants that serve market (n,k)	(1)
T_g, $T_{g,n}$	plants of group g that serve a given market	Section 1.2
Households and demand		
L_n	labor supplied by country n	Agents
U_n, C_{nk}	utility; consumption of the sector-k bundle	(1)
β_k	share of income spent on sector k	(1)
σ_k	elasticity of substitution between varieties of sector k	(1)
$q_{i,n}$, $p_{i,n}$	quantity and price of plant i's variety in country n	(2)
P_{nk}	price index of sector k in country n	(2)
P_n	consumer price index of country n	Representative household
X_n	income of country n	(5)
E_{nk}	spending on market (n,k), the size of the market	(12)
Technology, trade and ownership		
w_ℓ	wage in country ℓ	(3)
φ_i	productivity of plant i	(3)
a_i	unit cost of plant i, w_ℓ/φ_i	(3)
$d_{\ell n}$	iceberg trade cost from ℓ to n (≥ 1)	(4)
$t_{\ell nk}$	tariff of country n on sector-k goods produced in ℓ	(4)
$c_{i,n}$	delivered marginal cost of plant i in market n, including the tariff	(4)
θ_{gn}	share of group g owned by country n	(5)
π_g, $\pi_{g,n}$	profit of group g, in total and in market $(n,k(g))$; within one market, π_g denotes $\pi_{g,n}$	(5)
Z	a proper subset of countries, in the condition for existence of wages	Existence and uniqueness
TR_n	tariff revenue of country n	(5)
Market structure (one market)		
A	market aggregate, $P^{1-\sigma}$; high when the market is competitive	(6)
s_i	revenue share of plant i in the market	(6)
S_g	revenue share of group g in the market	(6)
$\mu(S)$	markup of a group with share S	(7)
K_g	strength of group g, $\sum_{i\in T_g} c_i^{1-\sigma}$	(8)
$\psi(S)$	share map: a group with strength K has the share S with $\psi(S) = K/A$	(9)
$m(S)$	Lerner index, $1 - 1/\mu(S)$	(11)
κ	curvature of profit in the share, $\sigma - 1$	(11)

Symbol	Meaning	Where
General equilibrium		
$V_{i,n'}$	cost of plant i in serving country n', all paid to labor	(13)
$r_{i,n'}$	revenue of plant i in country n'	(13)
W_n	wage bill of country n (labor demanded, in value)	Section 1.3
M_{nj}	elasticity of W_n with respect to the wage of country j	Section 1.3
$b^n_{i,n'}$	share of W_n paid by plant i out of its sales in n'	(14)
$\bar{E}^{(j)}_n$	average response of the markets where country n's plants sell to w_j	(14)
D	set of wage vectors on which uniqueness is checked	Theorem 1
λ	damping of the wage update	Solution algorithm
z_n	excess demand for labor, $W_n/w_n - L_n$	Appendix B.2
Γ, τ_{nk}	income matrix; tariff revenue per dollar spent in (n,k)	Appendix B.1
$\hat{x}$	log change of x	Appendix B.3
$\hat{c}_g$	sales-weighted average change in group g's delivered costs	Appendix B.3
$\varepsilon_\mu(S)$	elasticity of the markup with respect to the share, $S/(1-S)$	Appendix B.3
Λ_g	how strongly group g's share responds to its relative cost	Appendix B.3
ϖ_g	weight of group g in the response of the price index	Appendix B.3
χ_g	coefficient of the between-group reallocation of labor payments	Appendix B.3
Extension: entry		
$F_{i,n}$	fixed cost for plant i to serve market n	(15)
f_k, $f^d_{\ell n}$	sector and bilateral components of the fixed cost	(15)
$K(T)$, $\Phi(T)$	strength and fixed cost of a set of plants T	(16)
$\Omega(K;A)$	profit of a group with strength K at aggregate A	(16)
$U_g(T;A)$	net payoff of group g from serving the market with T	(16)